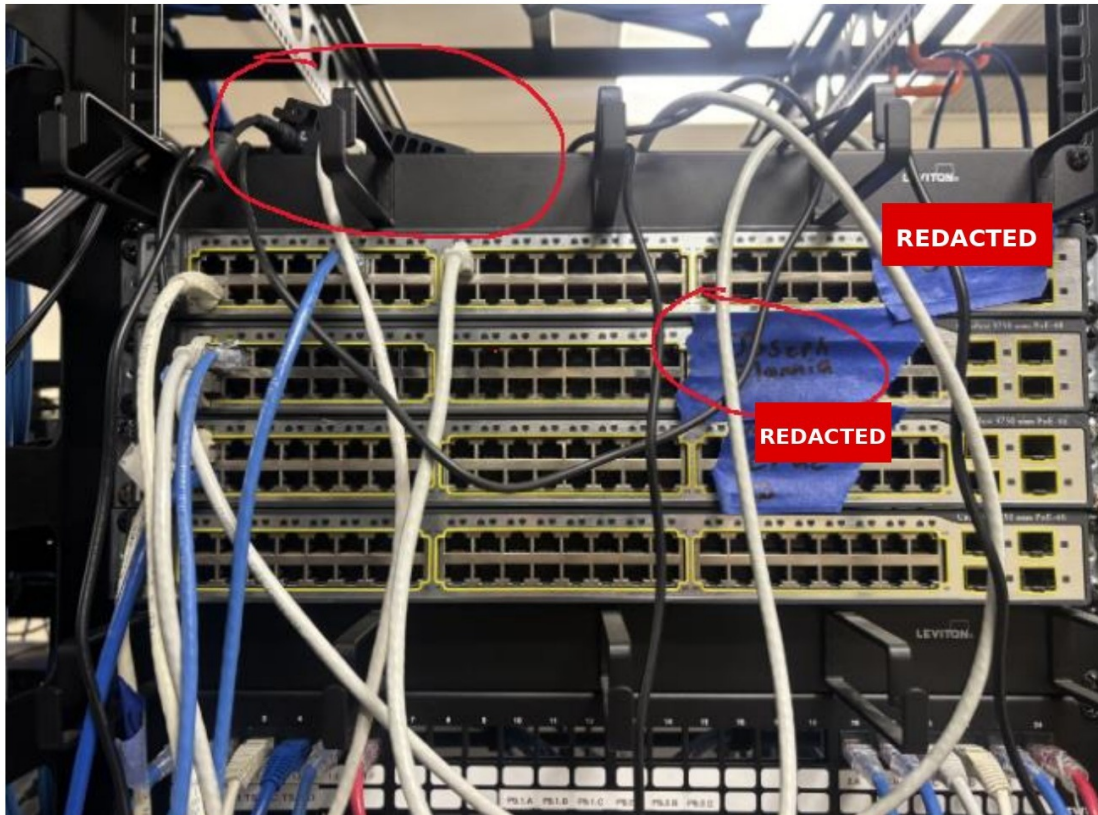


Part 1:

The first part of this course includes setting up initial devices, these being a NUC and a layer 3 switch. Any credentials, IP addressing, new devices, etc. Will be added directly to the network documentation spreadsheet.

The subnet as of now will be in the “10.18.x.x” format. This will change in part 2.

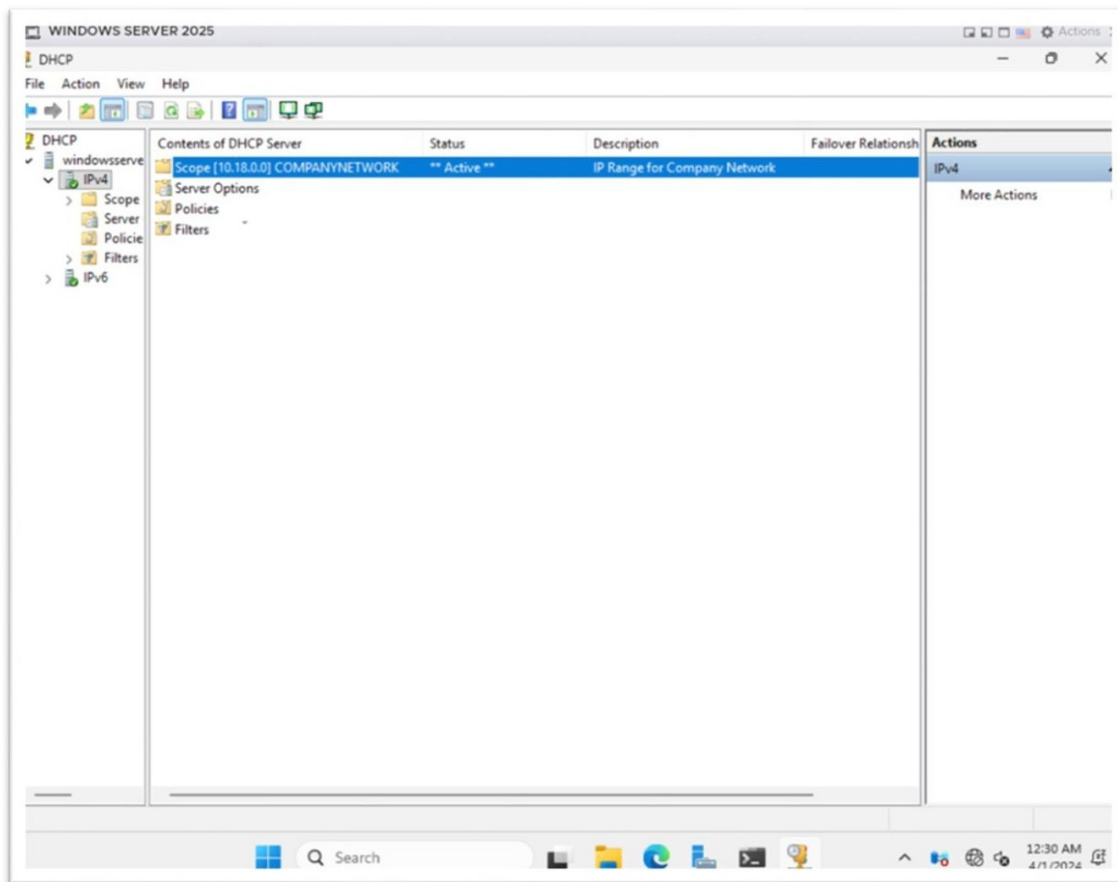
The physical devices are depicted below and are labeled accordingly:



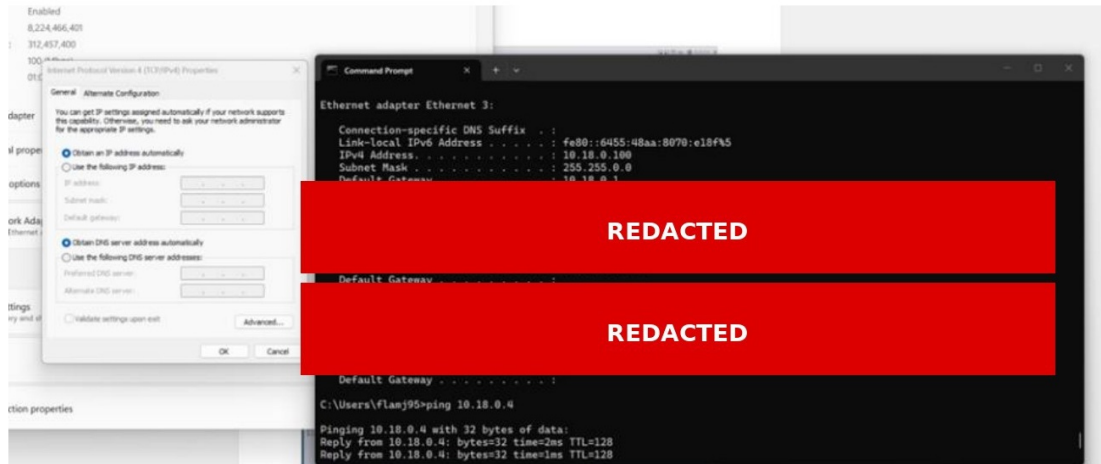
As of now, the only configurations required would be basic ones, such as IP addressing, to ensure connectivity.

After the NUC has been assigned a proper IP and installed with VMWare ESXI, a Windows 2025 Server was installed. After configuring a proper IP, DHCP was installed on the server.

Below is a screenshot proving DHCP is functional and installed on the server:



Classroom Computer receiving IP:



With the physical devices set up and the DHCP server functional, part one is complete. As stated before, all IP addressing, credentials, etc. Can be found in the documentation spreadsheet.

Part 2:

Part 2 of this course involves VLAN configurations, DHCP scopes, and routing.

On the switch, the following VLANs were created:

- **VLAN10: Client-Computers**
- **VLAN20: Virtual-Clients**
- **VLAN30: Virtual-Servers**
- **VLAN40: Physical-Devices**
- **VLAN50: Parking-Lot**

VLAN	Name	Status	Ports
1	default	active	REDACTED
10	Client-Computers	active	
20	Virtual-Clients	active	
30	Virtual-Servers	active	
VLAN	Name	Status	Ports
40	Physical-Devices	active	
50	Parking-Lot	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

LAYER3SWITCH(config-vlan)#

The switch is connected to the NUC via interface **REDACTED**. Since multiple VLANs will have traffic going over this interface, it needs to be configured as a trunk port. Below are the commands entered to enable trunking.


```

LAYER3SWITCH(config)#interf REDACTED
LAYER3SWITCH(config-if)#description Link to NUC
LAYER3SWITCH(config-if)#swit
LAYER3SWITCH(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be co
nfigured to "trunk" mode.
LAYER3SWITCH(config-if)#switchport trunk enca
LAYER3SWITCH(config-if)#switchport trunk encapsulation dot1q
LAYER3SWITCH(config-if)#switchport mode trunk
LAYER3SWITCH(config-if)#switch
*Mar  7 22:58:07.970: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et2/0/2, changed state to downport t
% Incomplete command.

LAYER3SWITCH(config-if)#swit
LAYER3SWITCH(config-if)#switchport
*Mar  7 22:58:11.007: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et2/0/2, changed state to uptrunk allowed vlan 10,20,30,40,50
LAYER3SWITCH(config-if)#spann
LAYER3SWITCH(config-if)#spanning-tree por
LAYER3SWITCH(config-if)#spanning-tree portfaa
LAYER3SWITCH(config-if)#spanning-tree portfas
LAYER3SWITCH(config-if)#spanning-tree portfast tr
LAYER3SWITCH(config-if)#spanning-tree portfast trunk
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

LAYER3SWITCH(config-if)#no shutdown
LAYER3SWITCH(config-if)#exit

```

Interface **REDACTED** connects the classroom PC to the switch. This will be configured as an access port with the following commands:

```

LAYER3SWITCH(config)#interf REDACTED
LAYER3SWITCH(config-if)#switchport mode access
LAYER3SWITCH(config-if)#switchport access vlan 10
LAYER3SWITCH(config-if)#descri
*Mar  7 22:59:12.026: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, cha
nged state to downption Classroom PC
LAYER3SWITCH(config-if)#spann
LAYER3SWITCH(config-if)#spanning-tree po
LAYER3SWITCH(config-if)#spanning-tree port
% Ambiguous command: "spanning-tree port"
LAYER3SWITCH(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet2/0/1 but will only
have effect when the interface is in a non-trunking mode.
LAYER3SWITCH(config-if)#

```

IP routing was enabled with the following command as well:

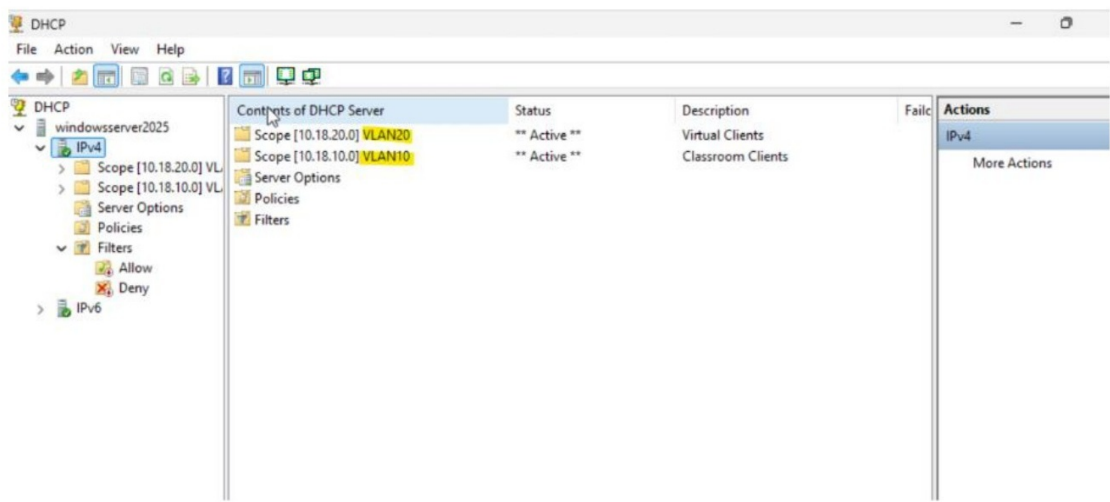
```

LAYER3SWITCH#
LAYER3SWITCH#
LAYER3SWITCH#ip routing

```

The Windows Server is now required to lease addresses to both client VLANs (10 and 20).

The scopes for both VLANs include addresses in the “10.18.10/20.50-200” range.



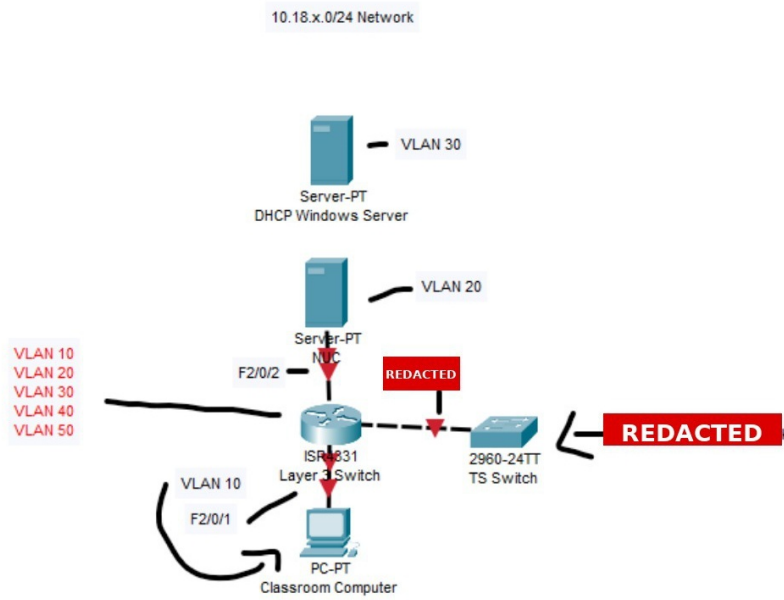
Port **REDACTED** on the switch is used for access to the TS Switch, with the following route entered:

```
LAYER3SWITCH(config)#ip route REDACTED
```

The site can be accessed via “**REDACTED IP**”.



As of now, this is a topology of the network in part 2.



Below is the classroom PC's output of the ipconfig /all command:

REDACTED

Below is the virtual client PC's output of the ipconfig /all command:

REDACTED

Below are pings from the classroom to the switch, TS switch, NUC, and Windows Server:

```
C:\Users\flamj95>ping 10.18.10.1

Pinging 10.18.10.1 with 32 bytes of data:
Reply from 10.18.10.1: bytes=32 time=2ms TTL=255
Reply from 10.18.10.1: bytes=32 time=2ms TTL=255
Reply from 10.18.10.1: bytes=32 time=2ms TTL=255
Reply from 10.18.10.1: bytes=32 time=2ms TTL=255

Ping statistics for 10.18.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

REDACTED

```
Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Users\flamj95>ping 10.18.40.2

Pinging 10.18.40.2 with 32 bytes of data:
Reply from 10.18.40.2: bytes=32 time=1ms TTL=63
Reply from 10.18.40.2: bytes=32 time=1ms TTL=63
Reply from 10.18.40.2: bytes=32 time=1ms TTL=63
Reply from 10.18.40.2: bytes=32 time=1ms TTL=63

Ping statistics for 10.18.40.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\flamj95>ping 10.18.30.3

Pinging 10.18.30.3 with 32 bytes of data:
Reply from 10.18.30.3: bytes=32 time=1ms TTL=127
Reply from 10.18.30.3: bytes=32 time=1ms TTL=127
Reply from 10.18.30.3: bytes=32 time=1ms TTL=127
Reply from 10.18.30.3: bytes=32 time=1ms TTL=127

Ping statistics for 10.18.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
```


Part 3:

Part 3 of this course focuses on switch configurations. The configurations include the following:

- Host Name
- SVI addresses in the Cisco Device Vlan

```
LAYER3SWITCH#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Vlan1          unassigned      YES manual up          down
Vlan10         10.18.10.1      YES manual up          up
Vlan20         10.18.20.1      YES manual up          up
Vlan30         10.18.30.1      YES manual up          up
Vlan40         10.18.40.1      YES manual up          up
Vlan50         10.18.50.1      YES manual up          up
```

- SSH Access

```
LAYER3SWITCH(config)#ip domain-name test.com
LAYER3SWITCH(config)#crypto key ge
LAYER3SWITCH(config)#crypto key generate r
LAYER3SWITCH(config)#crypto key generate rsa
% You already have RSA keys defined named LAYER3SWITCH.test.com.
% Do you really want to replace them? [yes/no]: n
```

```
line vty 0 4
transport input ssh
line vty 5 15
transport input ssh
```

- Secure all unused ports

```
REDACTED
LAYER3SWITCH(config-if-range)#shutdown
```

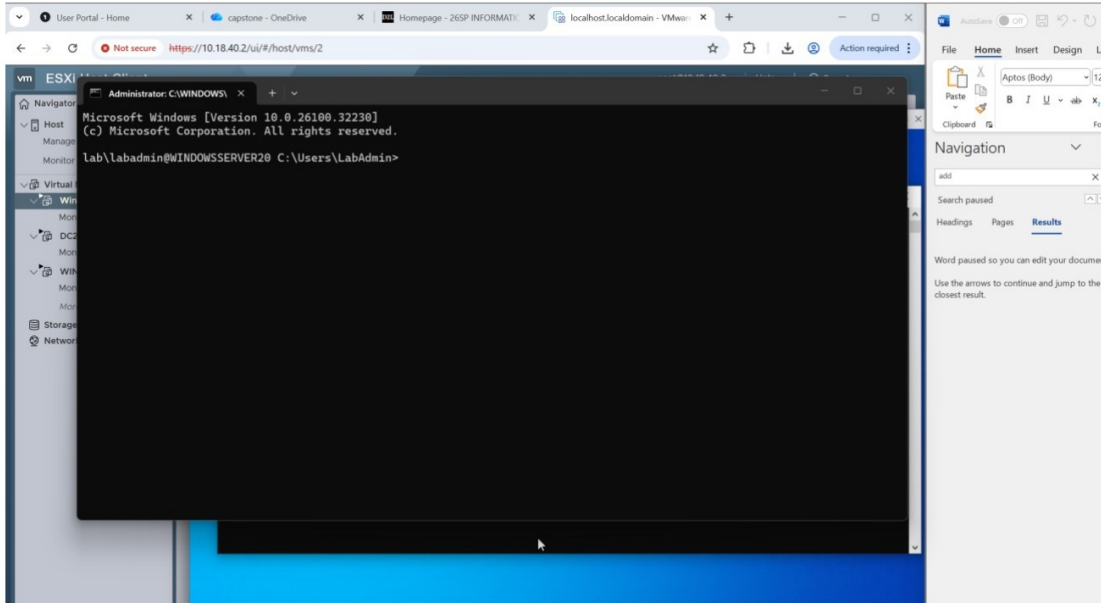
- MOTD banners

```
LAYER3SWITCH(config-if-range)#banner motd "Unauthorized Users will be Trespass$"
```

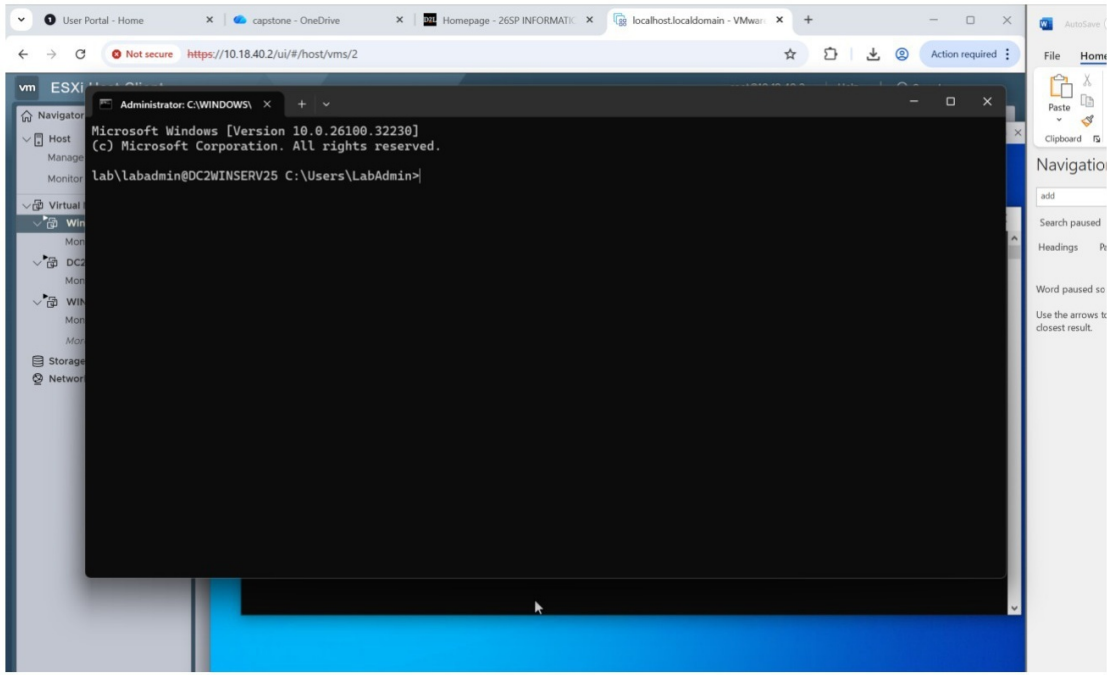
Below is proof of connectivity between devices using RDP and SSH.

SSH and RDP to Windows Server from Classroom PC:

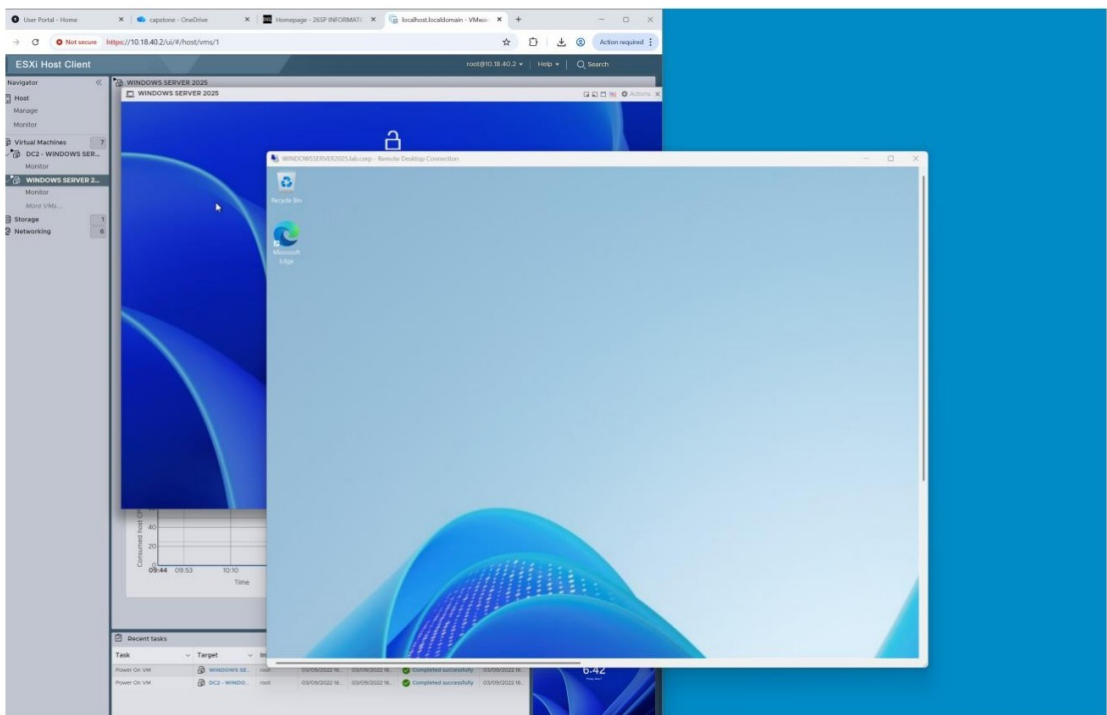
DC1



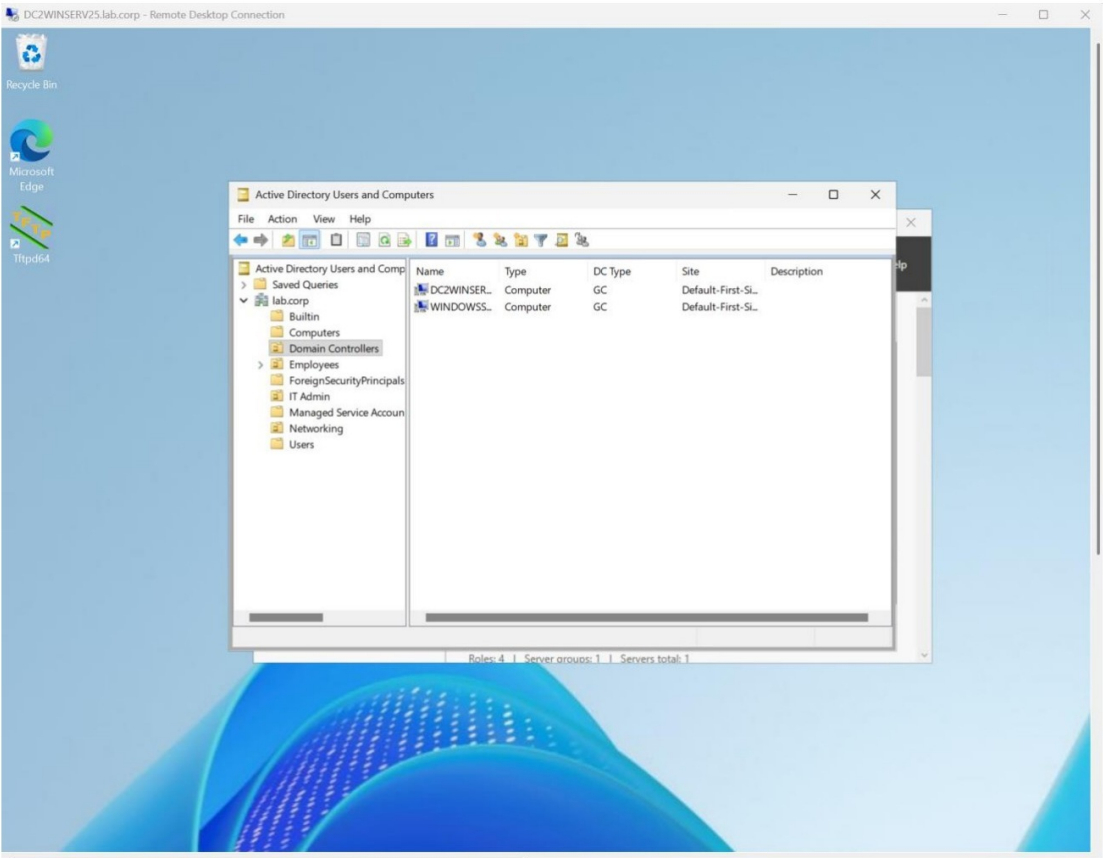
DC2



DC1

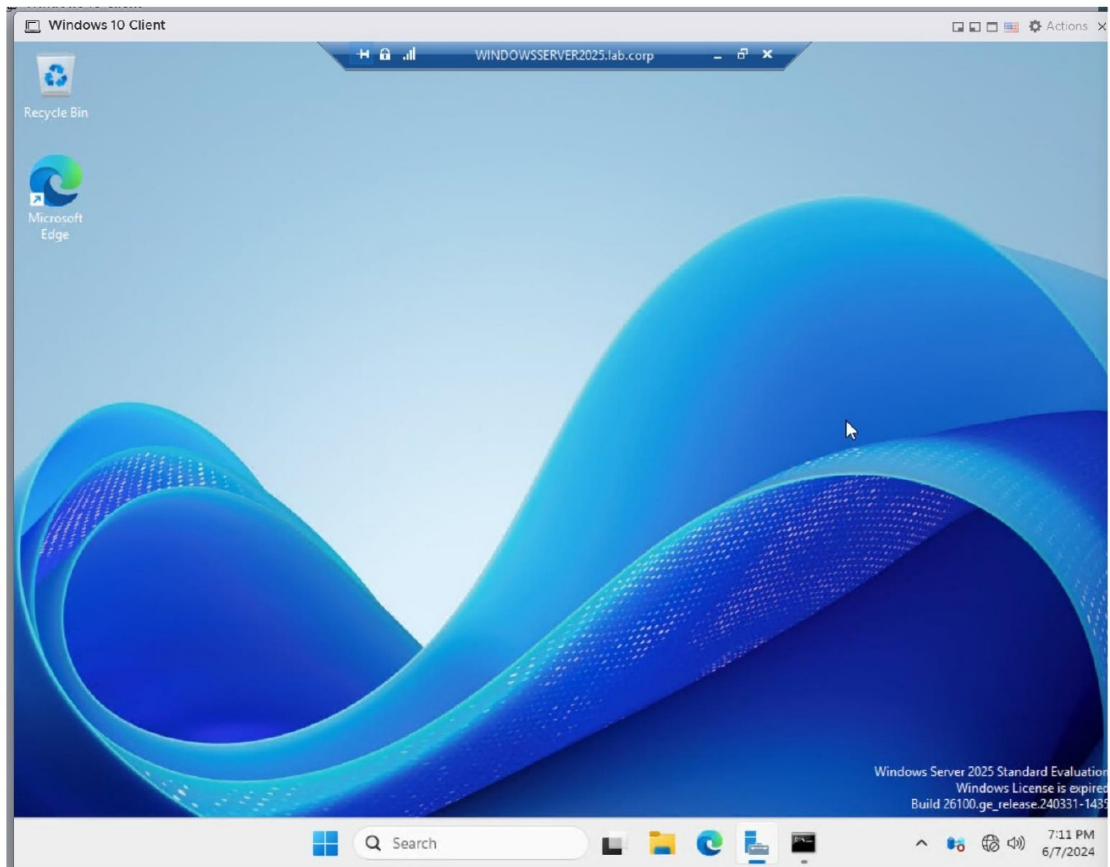


DC2

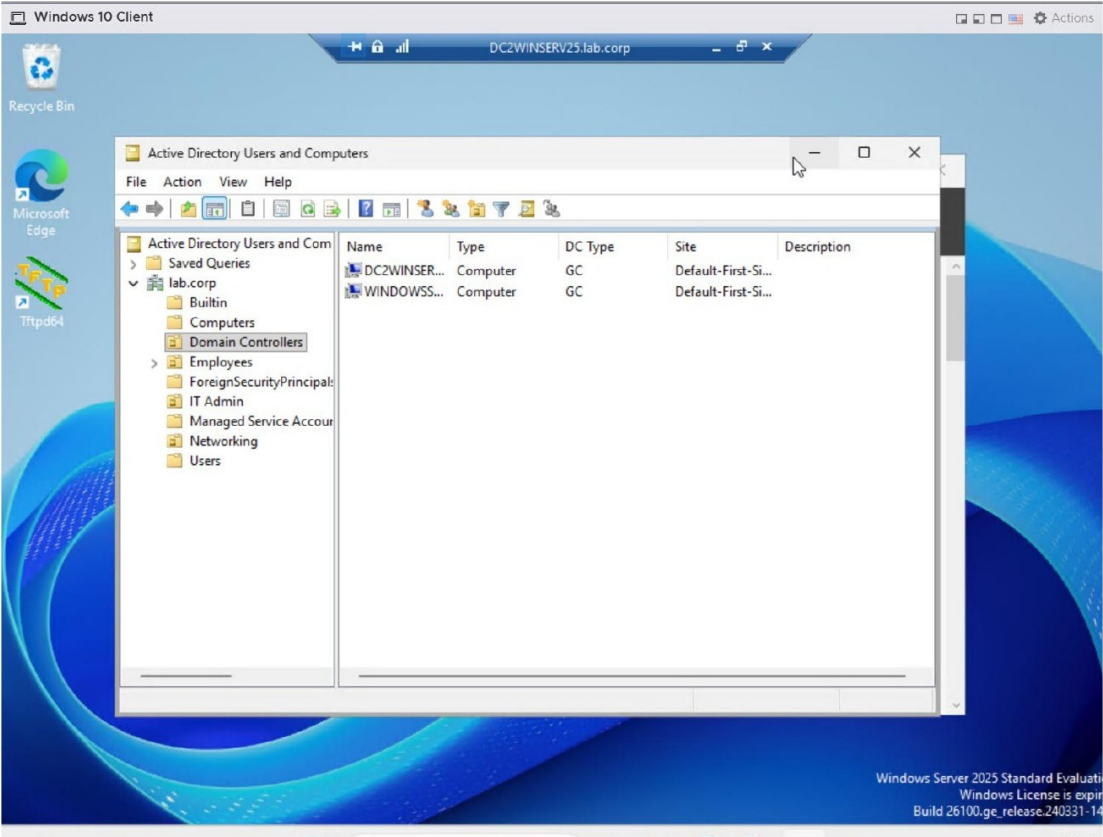


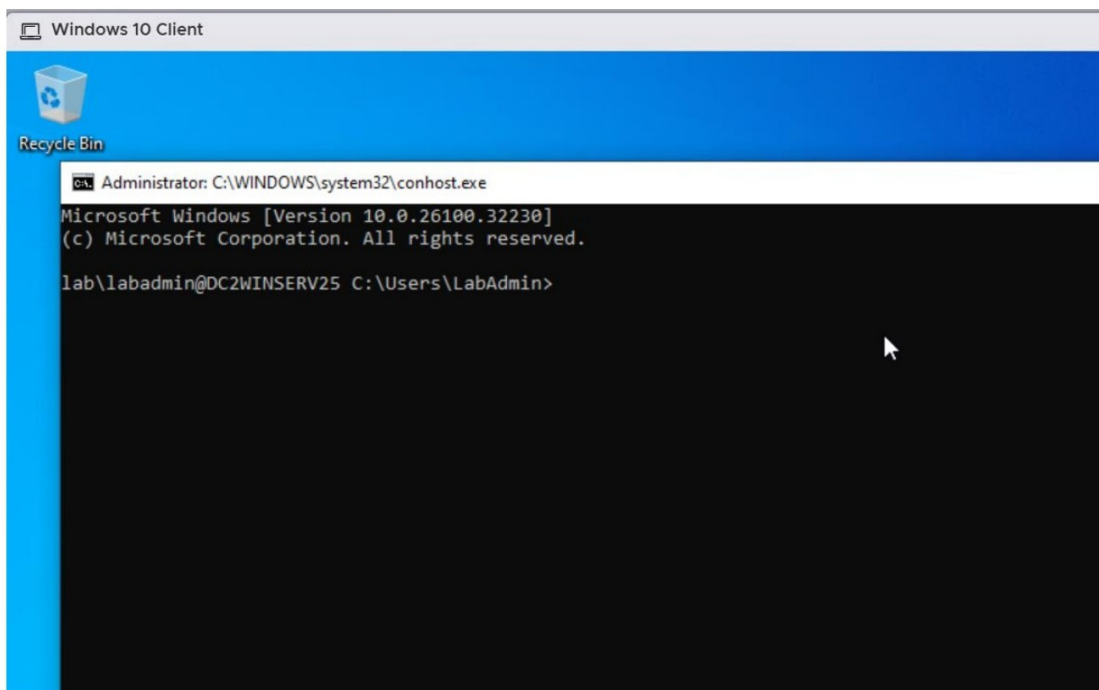
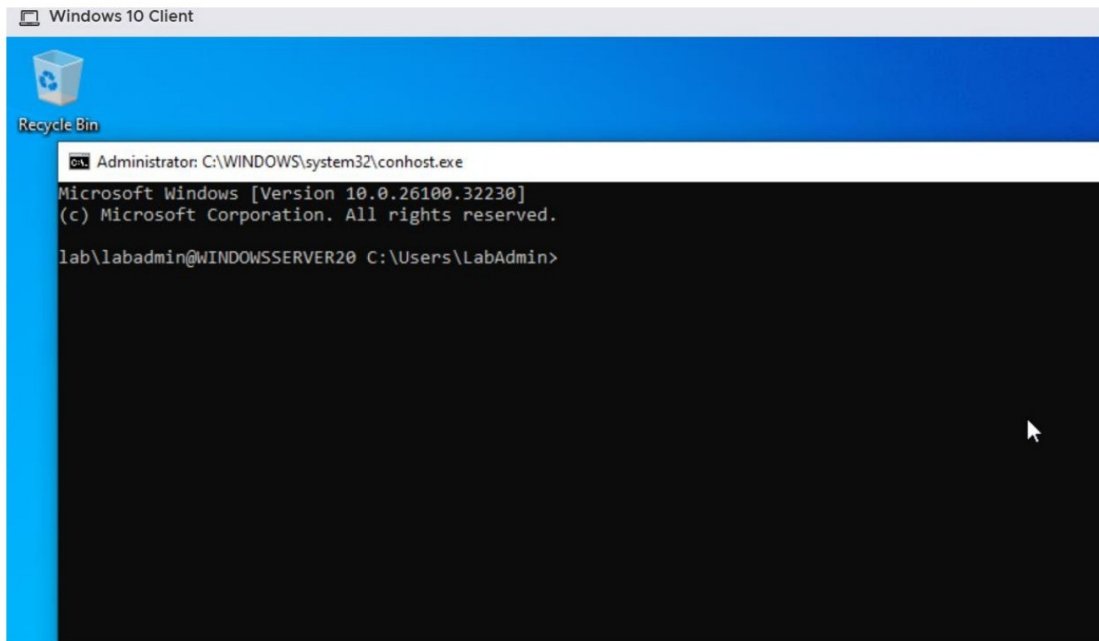
SSH and RDP to Windows Server from Virtual Client PC:

DC1



DC2





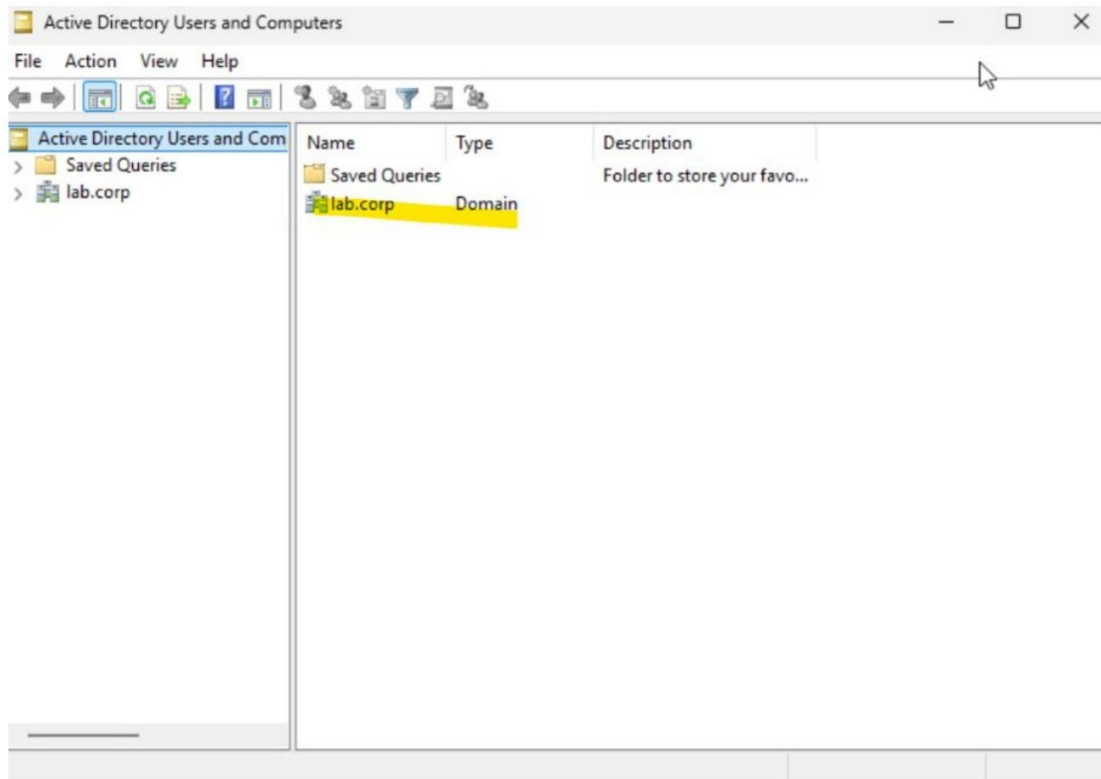
SSH from Windows server to switch:

```
C:\Windows\System32>ssh -oKexAlgorithms+=diffie-hellman-group1-sha1 -oHostKeyAlgorithms+=ssh-rsa -oCiphers+=aes128-cbc
oMACs+=hmac-sha1 wwwhite@10.18.30.1
The authenticity of host '10.18.30.1 (10.18.30.1)' can't be established.
RSA key fingerprint is SHA256:PmAbJlhynrSIRoalumZWsAkvrKsQ1sMD9SzoMr129kw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.18.30.1' (RSA) to the list of known hosts.
(wwwhite@10.18.30.1) Password:
Unauthorized Users will be Trespassed.
LAYER3SWITCH#
```

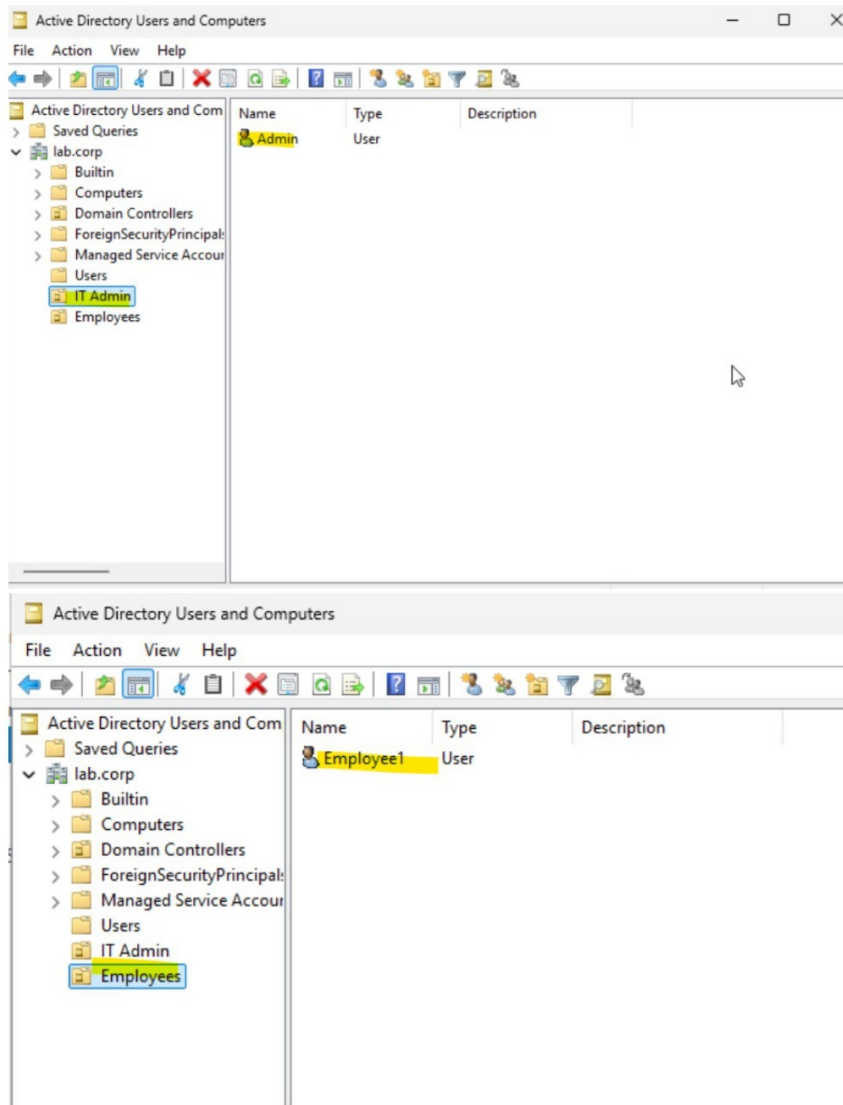

Part 4:

Part 4 of the course includes Active Directory setup and more deployed VMs.

Active Directory was installed on the DHCP server, and the domain name is **“lab.corp”**.

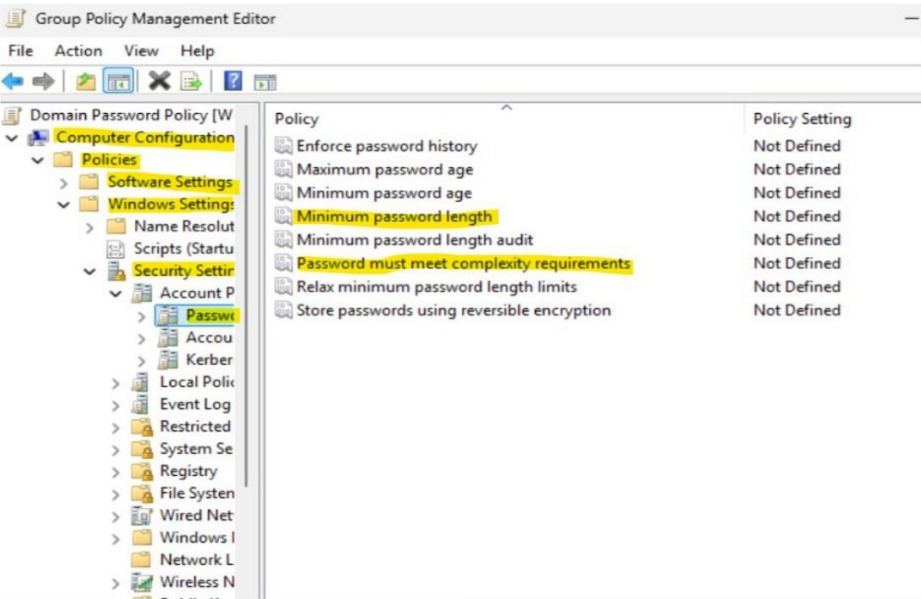
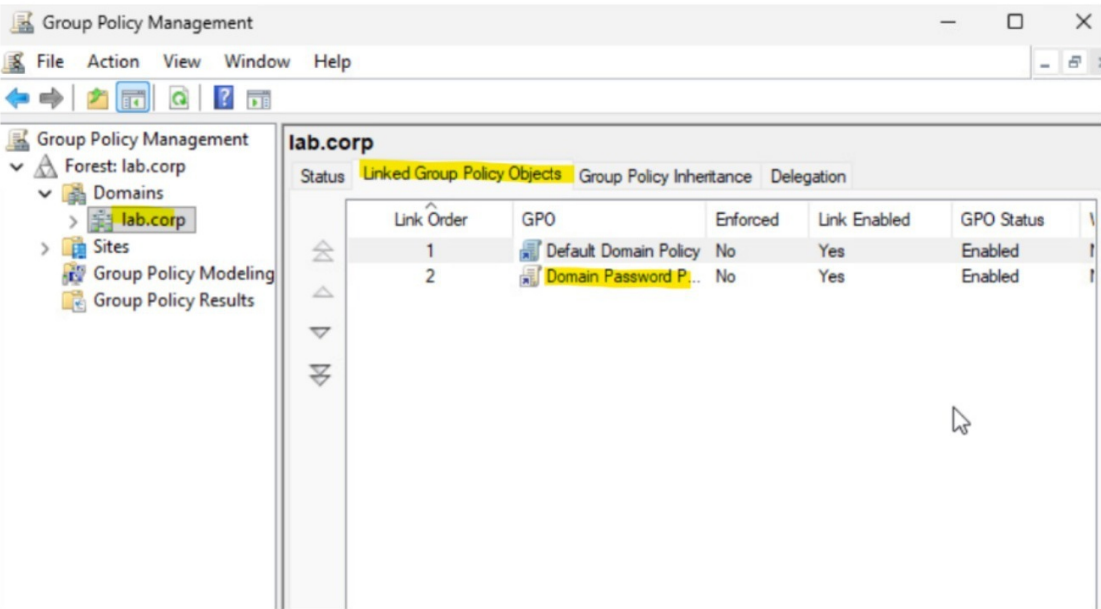


Two OUs were created, with an administrative and general user in the respective OUs.



A GP was created for the domain to enforce the following:

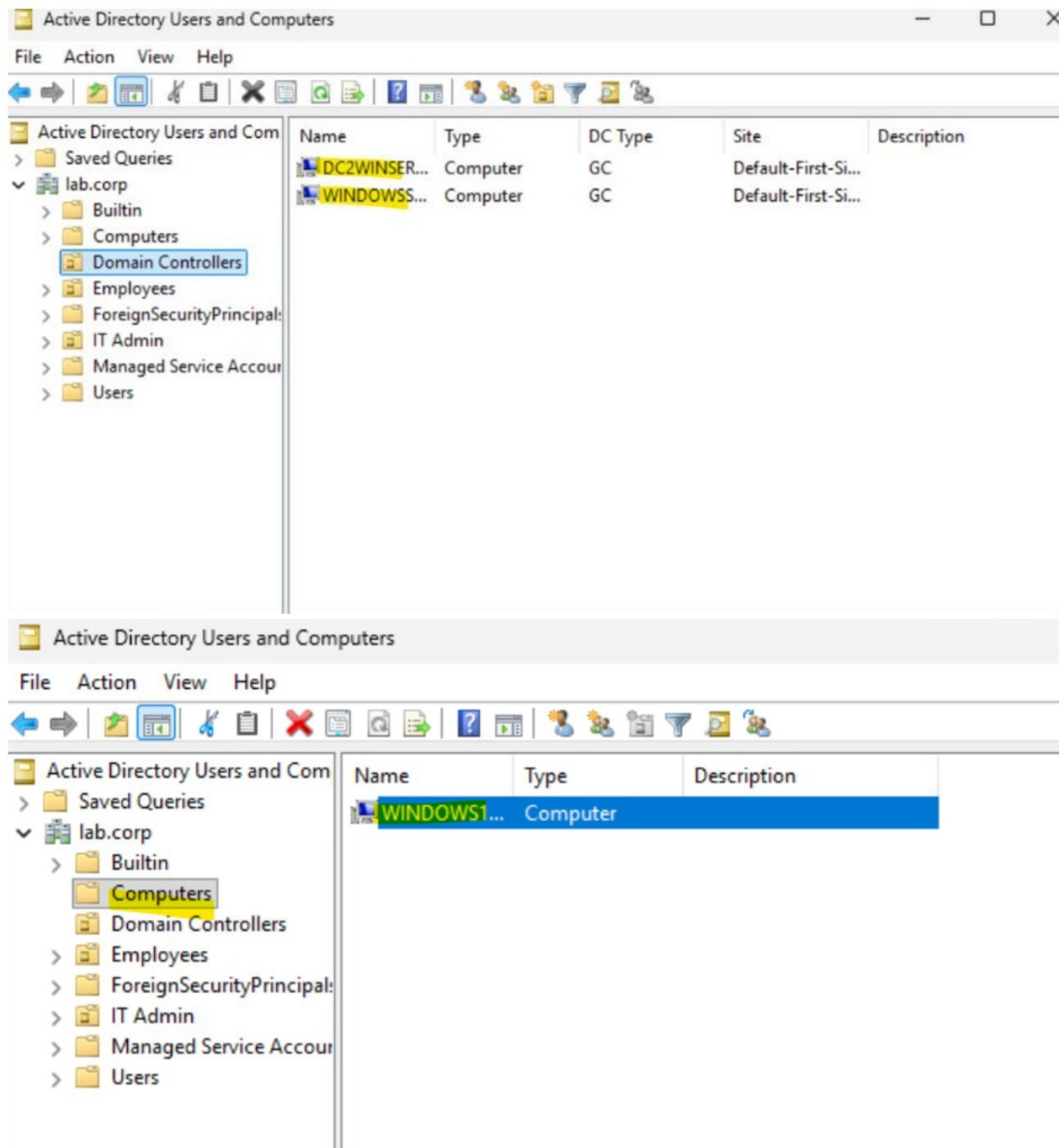
Password complexity and requires 10 character passwords.



Static DNS entries were created for all non-windows devices:

Name	Type	Data	Timestamp
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[21], windowsserver2025.l...	static
(same as parent folder)	Name Server (NS)	windowsserver2025.lab.co...	static
(same as parent folder)	Host (A)	10.18.30.3	4/12/202
windowsserver2025	Host (A)	10.18.30.3	static
NUC	Host (A)	10.18.40.2	
L3SWITCH	Host (A)	10.18.30.1	

A second DC and a Windows 10 client were also added to the environment:

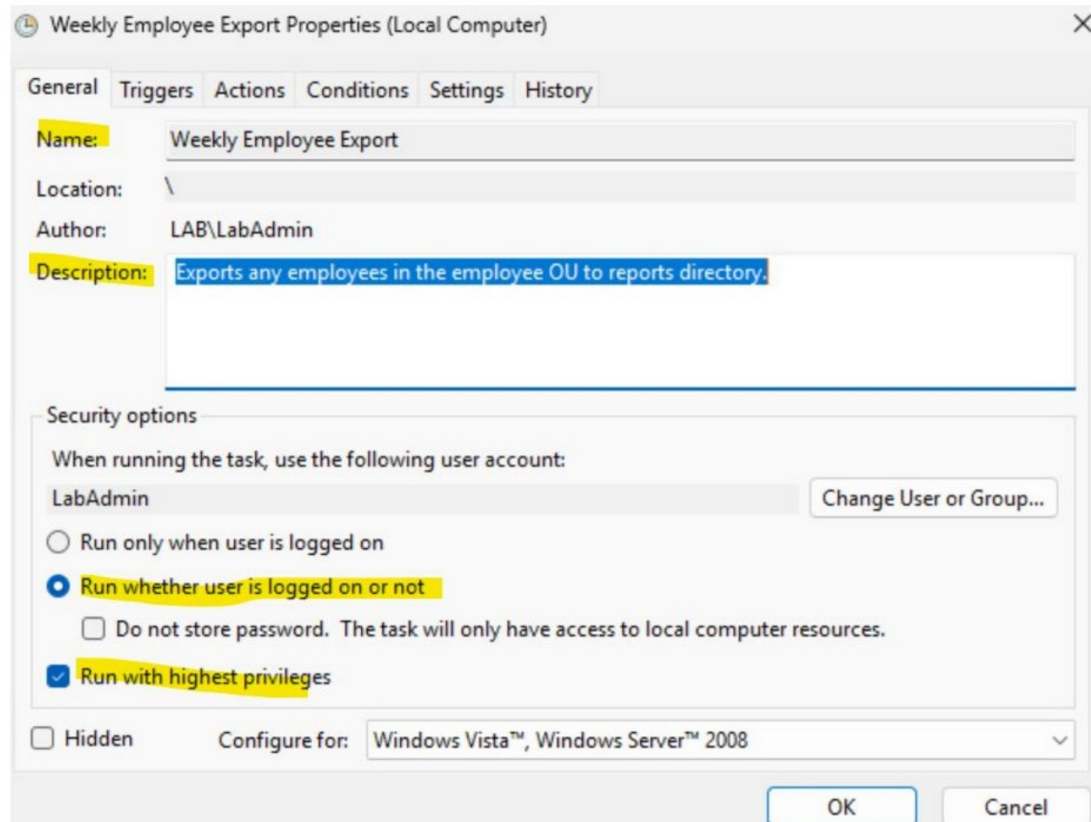


A PowerShell script was created to help with automation and ease of tasks.

The script below will export any users in the Employee OU to a text document called "EmployeeList.txt".

"Get-ADUser -Filter * -SearchBase "OU=Employees,DC=lab,DC=local" -Properties DisplayName,Department |Select-Object Name,SamAccountName,Department |Out-File "C:\Reports\EmployeeList.txt"

Task Manager was used to create a task to allow this script to run weekly.



Output of script:

EmployeeList.txt - Notepad		
File Edit Format View Help		
Name	SamAccountName	Department

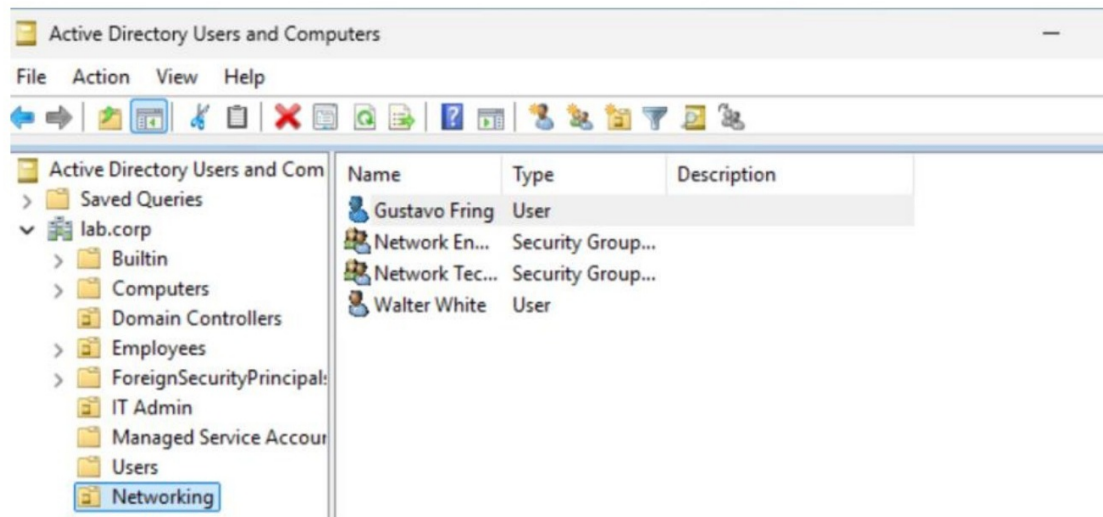
REDACTED	663452	
	664723	
	668793	
	675746	
	681397	
	686217	
	691530	
	705717	
	707501	
	714013	
	718006	
	720826	
	729630	
	738523	
	740019	
	740347	
	753994	
	761473	
	769980	
	773199	
	779504	
	802272	
	806170	
	811392	
	812335	
	822330	
	830999	
	839278	
	839796	
	848774	
	848807	

Part 5:

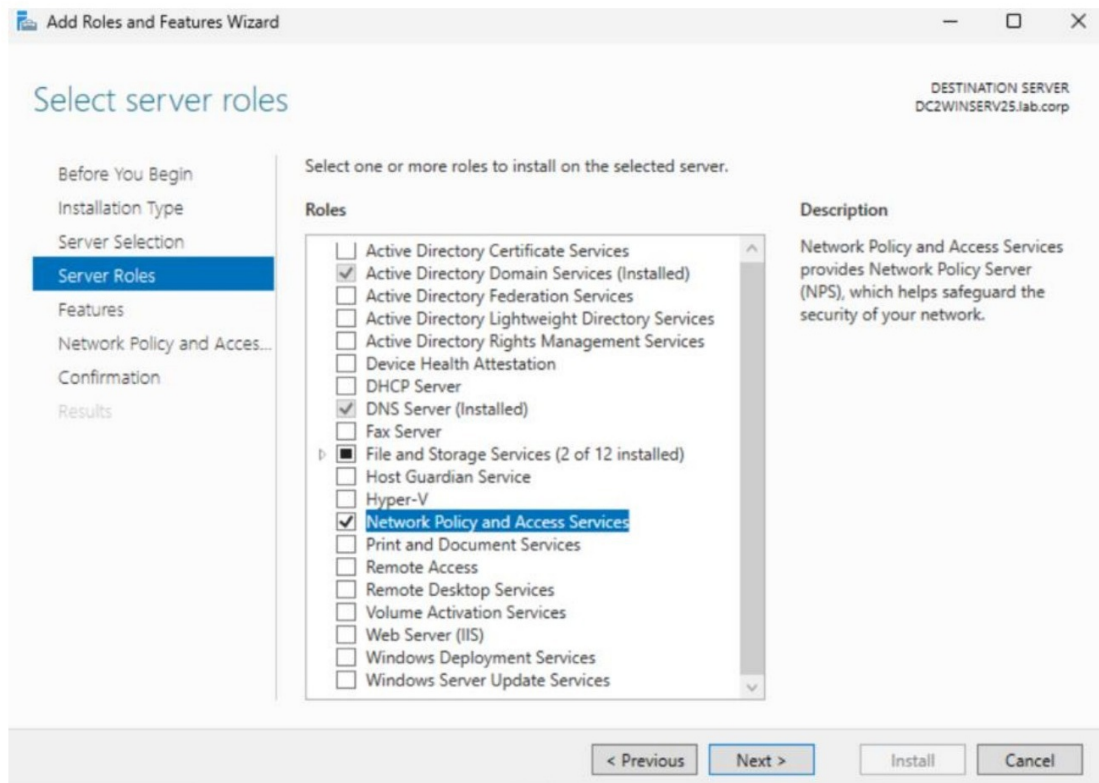
Part 5 of the course includes RADIUS configurations and backups.

The following OU, groups, and users were made to test RADIUS:

Walter White belongs to the Network Engineer group, and Gustavo Fring belongs to the Network Tech group.



RADIUS was installed on the second domain controller.



Configurations for RADIUS are mainly done in the Network Policy Server application.

Users in the engineer group should be able to log into the switch with full access, and users in the network group should only have read-only access. This can be configured via a new network policy.

New Network Policy ✕



Specify Network Policy Name and Connection Type

You can specify a name for your network policy and the type of connections to which the policy is applied.

Policy name:

Network connection method
Select the type of network access server that sends the connection request to NPS. You can select either the network access server type or Vendor specific, but neither is required. If your network access server is an 802.1X authenticating switch or wireless access point, select Unspecified.

☒ **Type of network access server:**

☐ **Vendor specific:**

Previous **Next** Finish Cancel

Shown below is the engineer and tech users connecting to the switch with correct privileges:

```
Microsoft Windows [Version 10.0.26100.32230]
(c) Microsoft Corporation. All rights reserved.

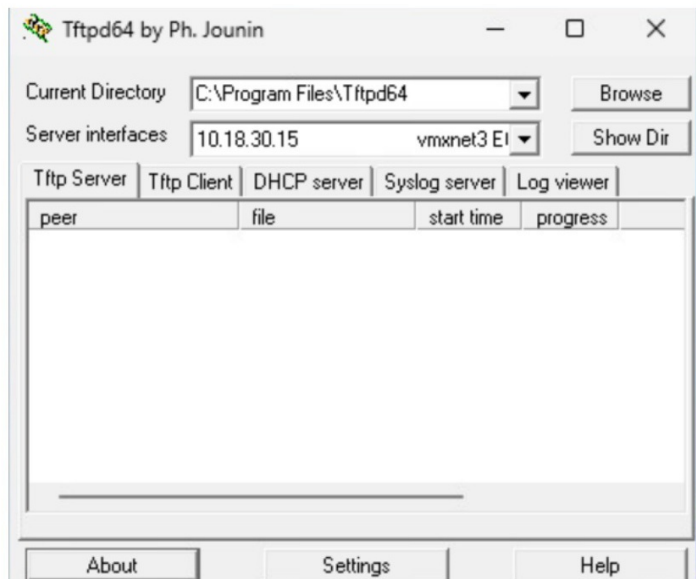
C:\Users\LabAdmin>ssh -oKexAlgorithms=+diffie-hellman-group1-sha1 -oHostKeyAlgorithms=+ssh-rsa -oCiphers=aes128-cbc -oMACs=+hmac-sha1 gfring@10.18.40.1
(gfring@10.18.40.1) Password:
(gfring@10.18.40.1) Password:
Unauthorized Users will be Trespassed.
LAYER3SWITCH>show priv
LAYER3SWITCH>show privilege
Current privilege level is 1

Command Prompt - ssh -oKe X + v

Microsoft Windows [Version 10.0.26100.32230]
(c) Microsoft Corporation. All rights reserved.

C:\Users\LabAdmin>ssh -oKexAlgorithms=+diffie-hellman-group1-sha1 -oHostKeyAlgorithms=+s
aes128-cbc -oMACs=+hmac-sha1 wwwhite@10.18.40.1
(wwwhite@10.18.40.1) Password:
Unauthorized Users will be Trespassed.
LAYER3SWITCH#show privilege
Current privilege level is 15
LAYER3SWITCH#
```

TFTP is used on DC2 to backup all network device configurations.



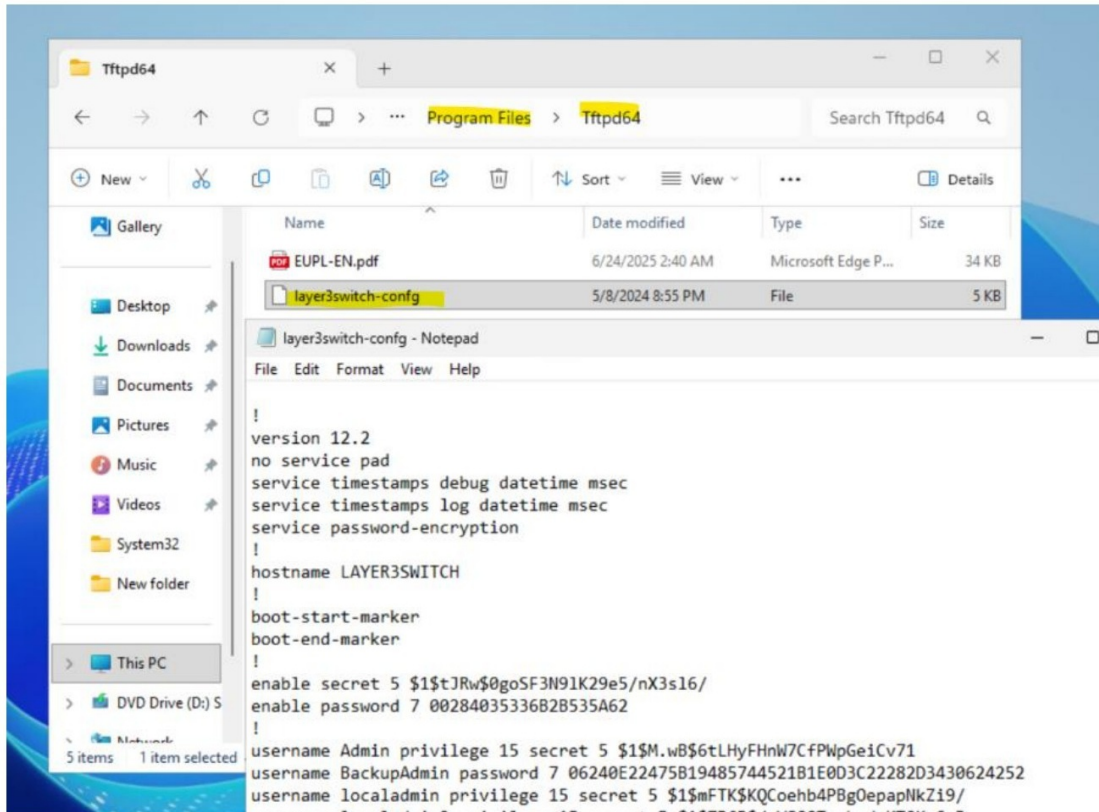
Backups can be made on the switch via the command below:

```

LAYER3SWITCH#copy running-config tftp
Address or name of remote host []? 10.18.30.15
Destination filename [layer3switch-config]?
!!
4322 bytes copied in 2.147 secs (2013 bytes/sec)
LAYER3SWITCH#

```

Backups are stored in the “Tftpd64” directory via a text file.



Below are PowerShell properties of the engineer and tech users:

Engineer:

```

Windows PowerShell
PS C:\WINDOWS\system32> Get-ADUser wwhite -Properties *

AccountExpirationDate      :
accountExpires             :
AccountLockoutTime         :
AuthenticationPolicy        : {}
AuthenticationPolicySilo   : {}
BadLogonCount              :
CannotChangePassword       : False
CanonicalName              :
Certificates                : {}
City                      :
CN                        : Walter White
codePage                   : 0
Company                   :
CompoundIdentitySupported   : {}
Country                   :
countryCode                : 0
Created                   :
Deleted                   :
Department                :
Description                :
DisplayName                : Walter White
DistinguishedName          : CN=Walter White,OU=Networking,DC=lab,DC=corp
Division                  :
EmailAddress               :
EmployeeID                :
EmployeeNumber             :
Fax                       :
GivenName                  : Walter
HomeDirectory              :
HomeDrive                  :
VirtualServerName          :
Organization                :
OtherName                  :
PasswordLastSet            :
POBox                      :
PostalCode                 :
PrimaryGroup                : CN=Domain Users,CN=Users,DC=lab,DC=corp
primaryGroupID             : 513
PrincipalsAllowedToDelegateToAccount : {}
ProfilePath                :
ProtectedFromAccidentalDeletion : False
SamAccountName             : wwhite
sAMAccountType             : 805306368
ScriptPath                 :
sDRightsEffective          : 0
ServicePrincipalNames      : {}
SID                        : S-1-5-21-1153590888-1499929391-3027196990-1605
SIDHistory                 : {}
sn                         : White
State                      :
StreetAddress               :
Surname                    : White
Title                      :
userCertificate             : {}
UserPrincipalName          : wwhite@lab.corp

```

Tech:

```
PS C:\WINDOWS\system32> Get-ADUser gfring -Properties *
```

```
AccountExpirationDate      :  
accountExpires             :  
AccountLockoutTime         :  
AuthenticationPolicy       : {}  
AuthenticationPolicySilo   : {}  
BadLogonCount              :  
CannotChangePassword       : False  
CanonicalName              :  
Certificates                : {}  
City                       :  
CN                          : Gustavo Fring  
codePage                   : 0  
Company                    :  
CompoundIdentitySupported  : {}  
Country                    :  
countryCode                : 0  
Created                    :  
Deleted                    :  
Department                 :  
Description                 :  
DisplayName                 : Gustavo Fring  
DistinguishedName           : CN=Gustavo Fring,OU=Networking,DC=lab,DC=corp  
Division                   :  
EmailAddress               :  
EmployeeID                 :  
EmployeeNumber             :  
Fax                        :  
GivenName                  : Gustavo  
HomeDirectory               :
```

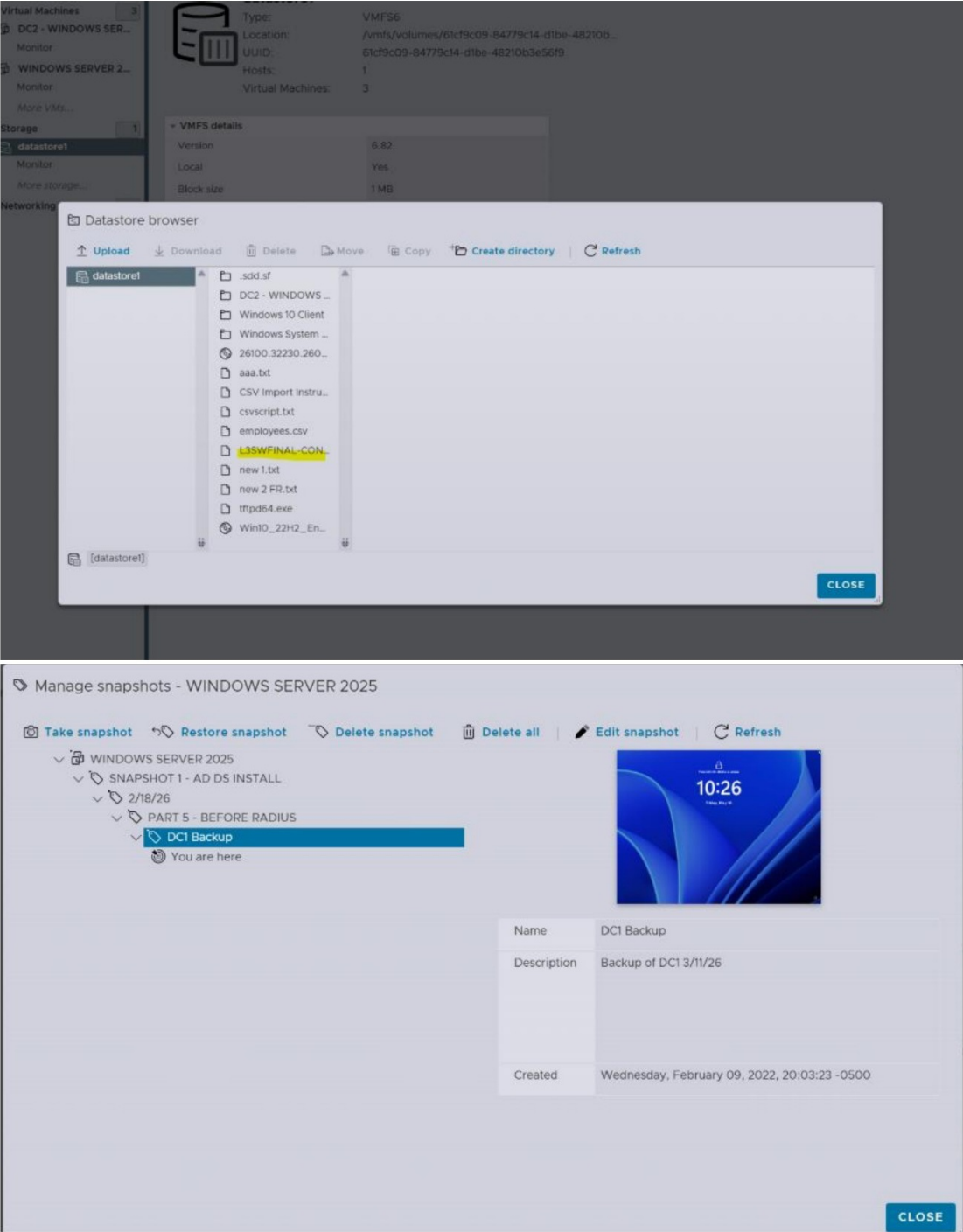
```
Country                    :  
countryCode                : 0  
Created                    :  
Deleted                    :  
Department                 :  
Description                 :  
DisplayName                 : Gustavo Fring  
DistinguishedName           : CN=Gustavo Fring,OU=Networking,DC=lab,DC=corp  
Division                   :  
EmailAddress               :  
EmployeeID                 :  
EmployeeNumber             :  
Fax                        :  
GivenName                  : Gustavo  
HomeDirectory               :  
HomeDrive                  :  
HomePage                   :  
HomePhone                  :  
Initials                   :  
instanceType               :  
isDeleted                  :  
KerberosEncryptionType     : {}  
LastBadPasswordAttempt     :  
LastKnownParent            :  
LastLogonDate              :  
LogonWorkstations          :  
Manager                    :  
MemberOf                   : {}  
MobilePhone                :  
Modified                   :  
Name                       : Gustavo Fring  
nTSecurityDescriptor        : System.DirectoryServices.ActiveDirectorySecurity  
ObjectCategory              : CN=Person,CN=Schema,CN=Configuration,DC=lab,DC=corp
```

```
Windows PowerShell
MemberOf : {}
MobilePhone :
Modified :
Name : Gustavo Fring
nTSecurityDescriptor : System.DirectoryServices.ActiveDirectorySecurity
ObjectCategory : CN=Person,CN=Schema,CN=Configuration,DC=lab,DC=corp
ObjectClass : user
ObjectGUID : d3b5e5ef-ce81-4d8a-9370-1b414a6c066e
objectSid : S-1-5-21-1153590888-1499929391-3027196990-1606
Office :
OfficePhone :
Organization :
OtherName :
PasswordLastSet :
POBox :
PostalCode :
PrimaryGroup : CN=Domain Users,CN=Users,DC=lab,DC=corp
primaryGroupID : 513
PrincipalsAllowedToDelegateToAccount : {}
ProfilePath :
ProtectedFromAccidentalDeletion : False
SamAccountName : gfring
sAMAccountType : 805306368
ScriptPath :
sDRightsEffective : 0
ServicePrincipalNames : {}
SID : S-1-5-21-1153590888-1499929391-3027196990-1606
SIDHistory : {}
sn : Fring
State :
StreetAddress :
Surname : Fring
Title :

SID : S-1-5-21-1153590888-1499929391-3027196990-1606
SIDHistory : {}
sn : Fring
State :
StreetAddress :
Surname : Fring
Title :
userCertificate : {}
UserPrincipalName : gfring@lab.corp

PS C:\WINDOWS\system32>
```

Backups of all VMs and network devices were also created via snapshots and the datastore browser on the NUC.



Manage snapshots - DC2 - WINDOWS SERVER 2025

[Take snapshot](#) | [Restore snapshot](#) | [Delete snapshot](#) | [Delete all](#) | [Edit snapshot](#) | [Refresh](#)

- ✓ [DC2 - WINDOWS SERVER 2025](#)
 - ✓ [PART 5 - BEFORE RADIUS](#)
 - ✓ [DC2 Backup](#)
 - [You are here](#)

[CLOSE](#)

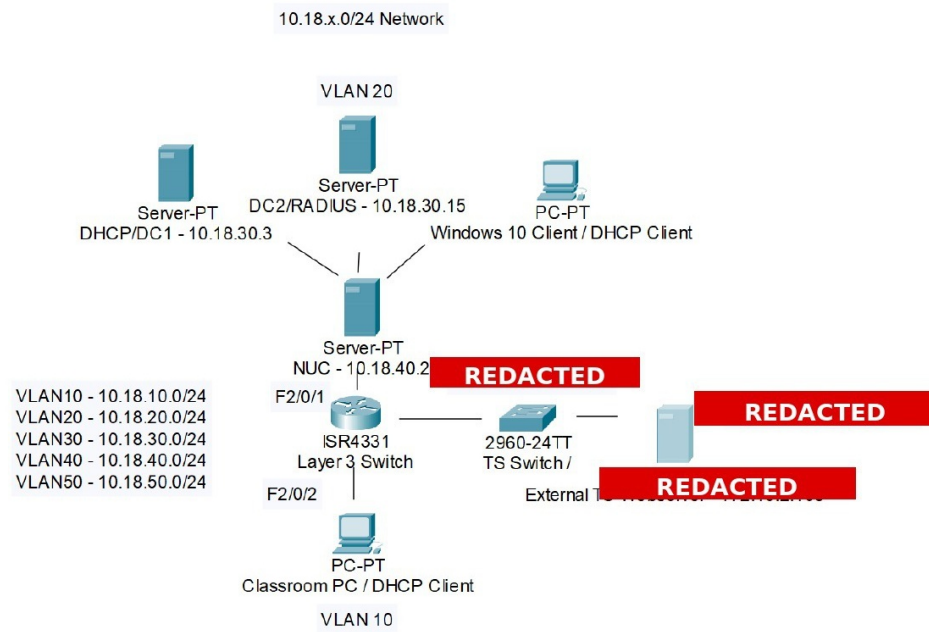
Manage snapshots - Windows 10 Client

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Below is a logical network diagram as of finishing part 5:



Below is the classroom PC pinging all devices depicted above.

```
lab\labadmin@WINDOWSSERVER20 C:\Users\LabAdmin>ping 10.18.10.1
```

```
Pinging 10.18.10.1 with 32 bytes of data:
```

```
Reply from 10.18.10.1: bytes=32 time=1ms TTL=255
```

```
Reply from 10.18.10.1: bytes=32 time=2ms TTL=255
```

```
Ping statistics for 10.18.10.1:
```

```
Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
```

Switch

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

```
Control-C
```

REDACTED

TS Switch

```
Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

```
Control-C
```

```
^C
```

```
lab\labadmin@WINDOWSSERVER20 C:\Users\LabAdmin>ping 10.18.40.2
```

NUC

```
Pinging 10.18.40.2 with 32 bytes of data:
```

```
Reply from 10.18.40.2: bytes=32 time<1ms TTL=63
```

```
Reply from 10.18.40.2: bytes=32 time<1ms TTL=63
```

```
Ping statistics for 10.18.40.2:
```

```
Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
Control-C
```

```
^C
```

```
lab\labadmin@WINDOWSSERVER20 C:\Users\LabAdmin>ping 10.18.30.3
```

DC1

```
Pinging 10.18.30.3 with 32 bytes of data:
```

```
Reply from 10.18.30.3: bytes=32 time<1ms TTL=128
```

```
Reply from 10.18.30.3: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 10.18.30.3:
```

```
Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
Control-C
```

```
^C
```

```
lab\labadmin@WINDOWSSERVER20 C:\Users\LabAdmin>ping 10.18.30.15
```

DC2

```
Pinging 10.18.30.15 with 32 bytes of data:
```

```
Reply from 10.18.30.15: bytes=32 time<1ms TTL=128
```

```
Reply from 10.18.30.15: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 10.18.30.15:
```

```
Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
Control-C
```

```
^C
```

```
lab\labadmin@WINDOWSSERVER20 C:\Users\LabAdmin>ping 10.18.20.5
```

Windows Client

```
Pinging 10.18.20.5 with 32 bytes of data:
```

```
Reply from 10.18.20.5: bytes=32 time<1ms TTL=127
```

```
Reply from 10.18.20.5: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 10.18.20.5:
```

```
Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```